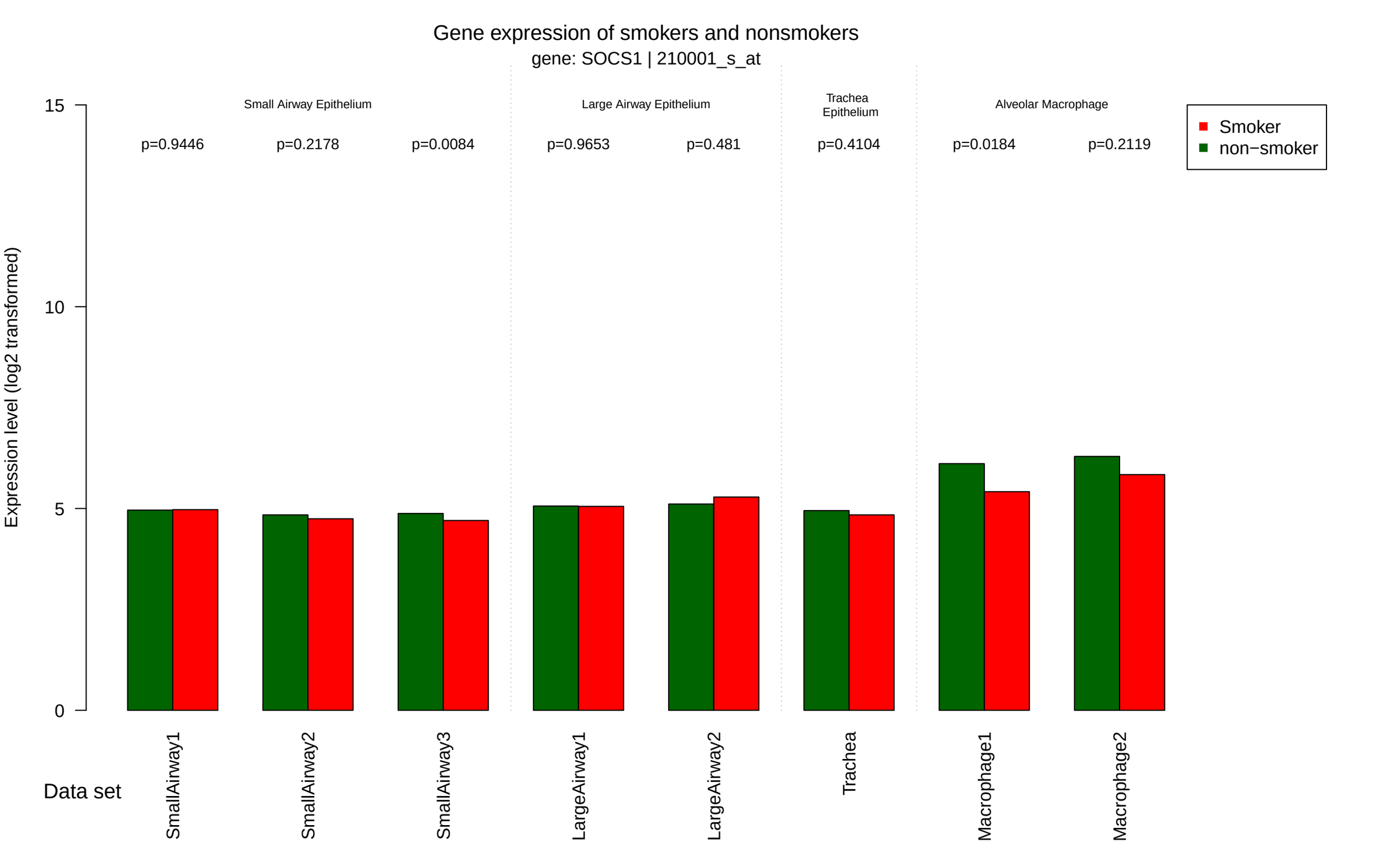
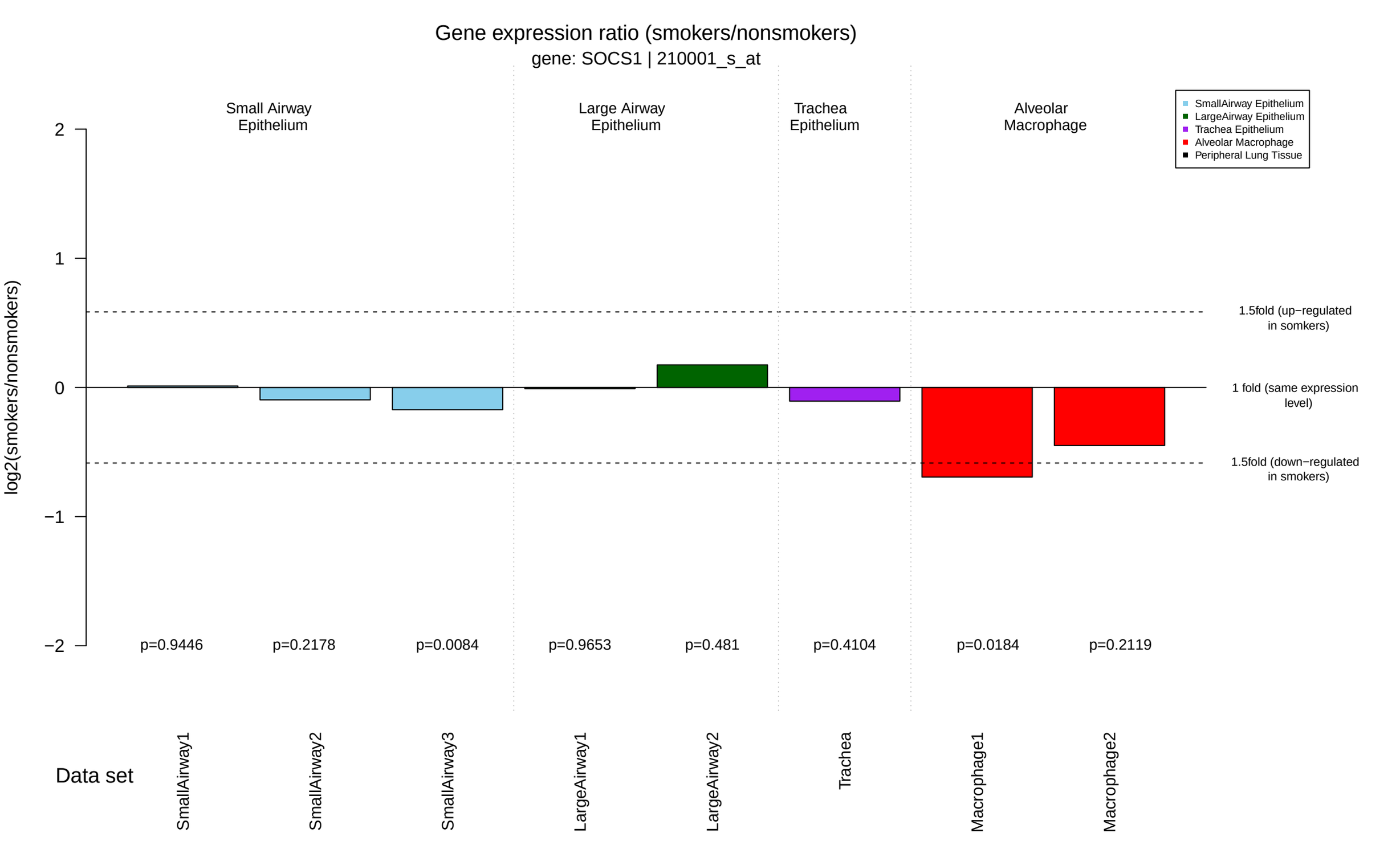
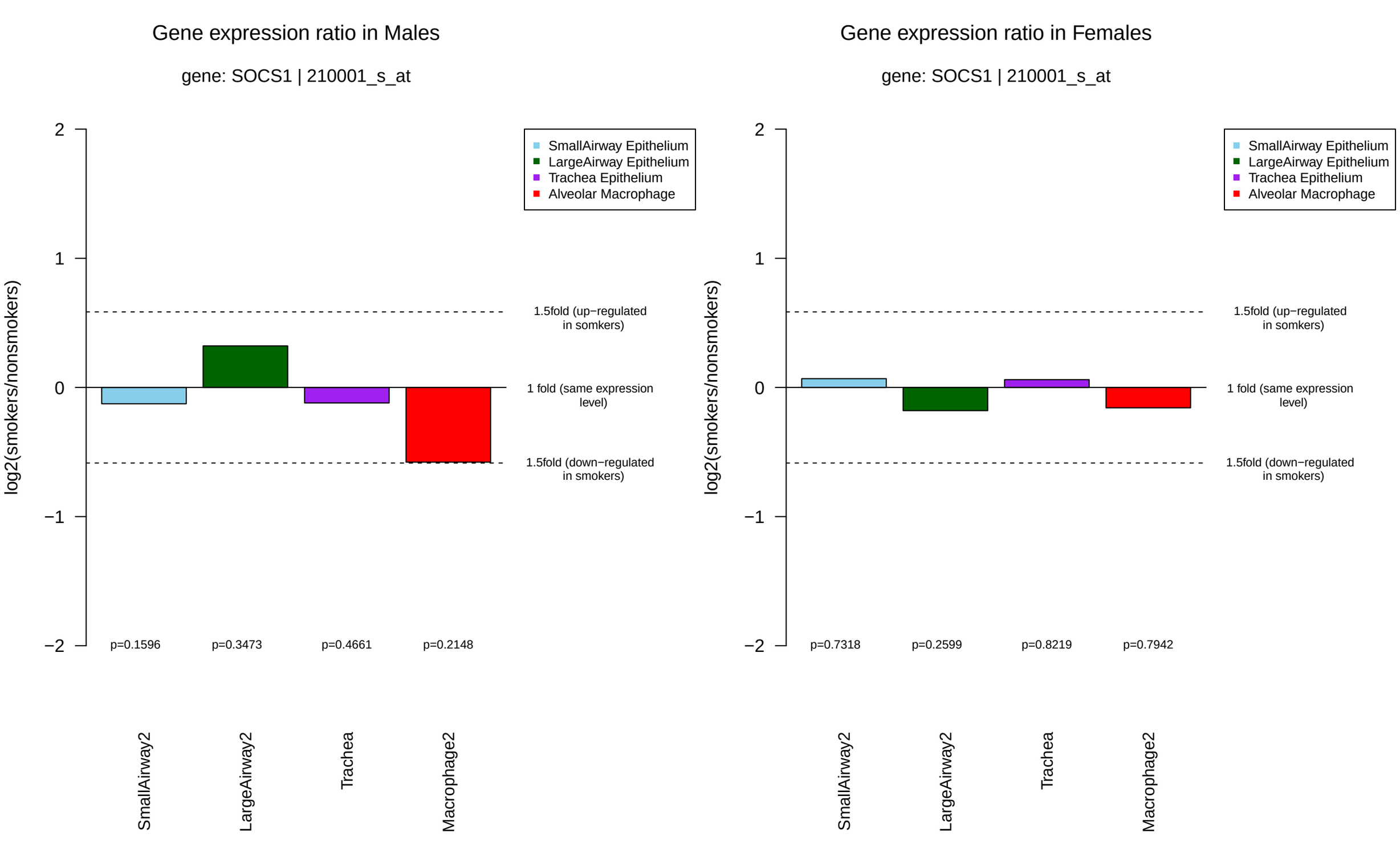
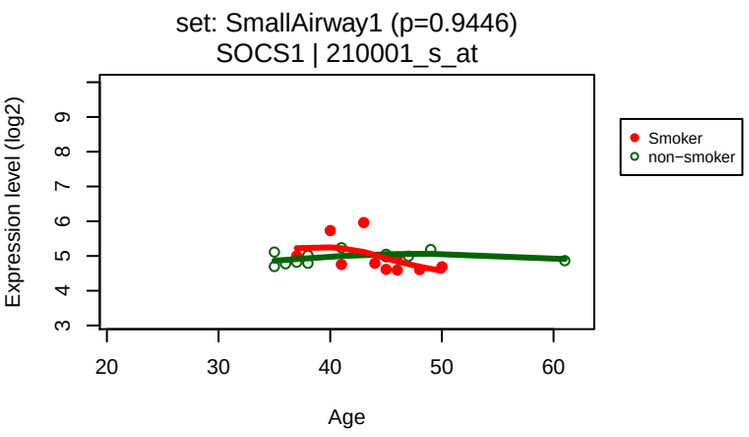
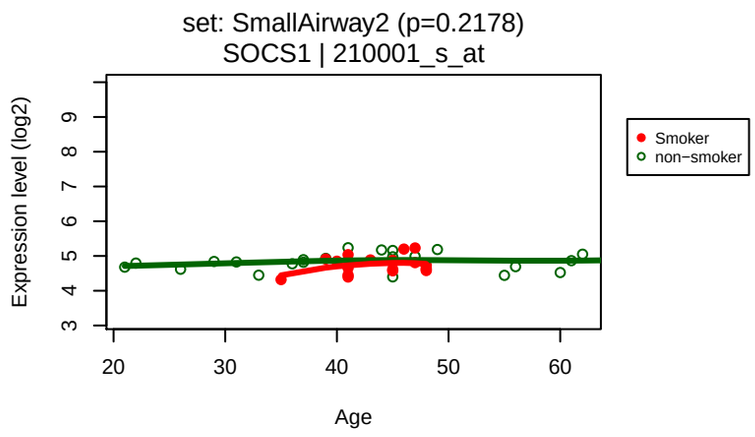
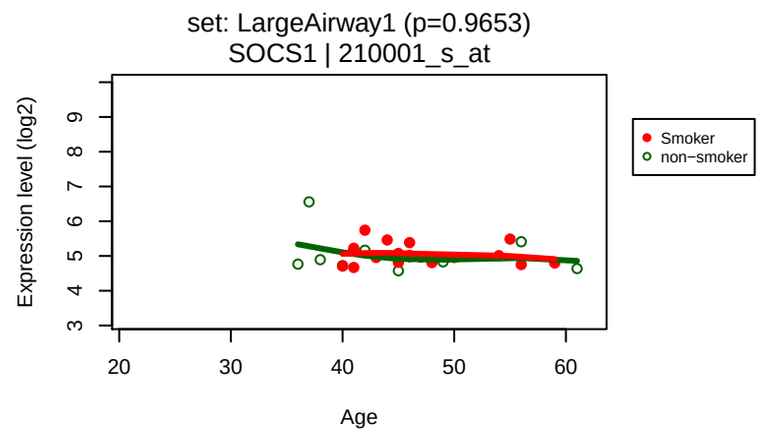

Gene expression of smokers and nonsmokers
gene: SOCS1 | 210001_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9446
p=0.2178
p=0.0084
p=0.9653
p=0.481
p=0.4104
p=0.0184
p=0.2119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

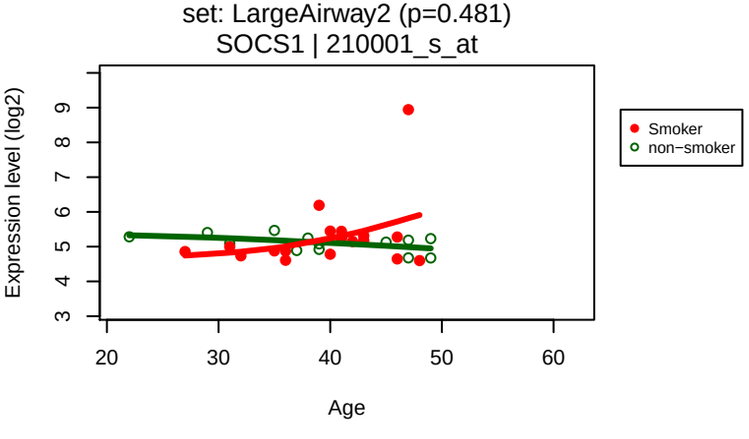

Gene expression ratio (smokers/nonsmokers)
gene: SOCS1 | 210001_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9446
p=0.2178
p=0.0084
p=0.9653
p=0.481
p=0.4104
p=0.0184
p=0.2119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

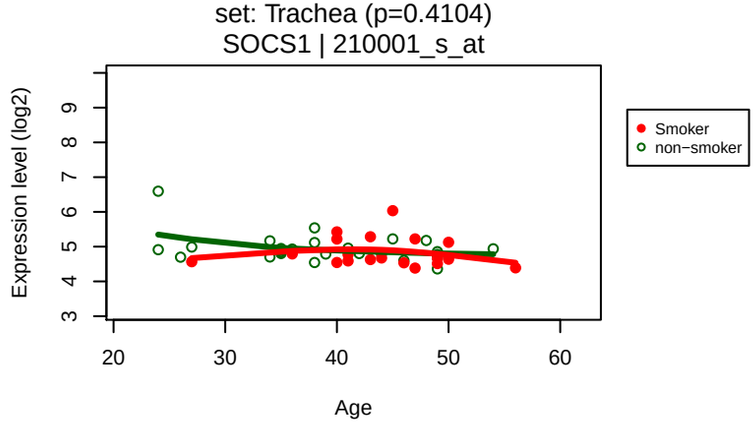

Gene expression ratio in Males
gene: SOCS1 | 210001_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1596
p=0.3473
p=0.4661
p=0.2148
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS1 | 210001_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7318
p=0.2599
p=0.8219
p=0.7942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

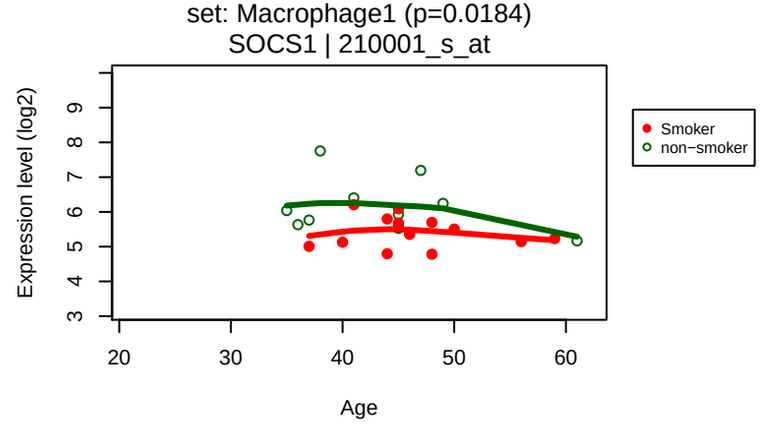

set: SmallAirway1 (p=0.9446)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker

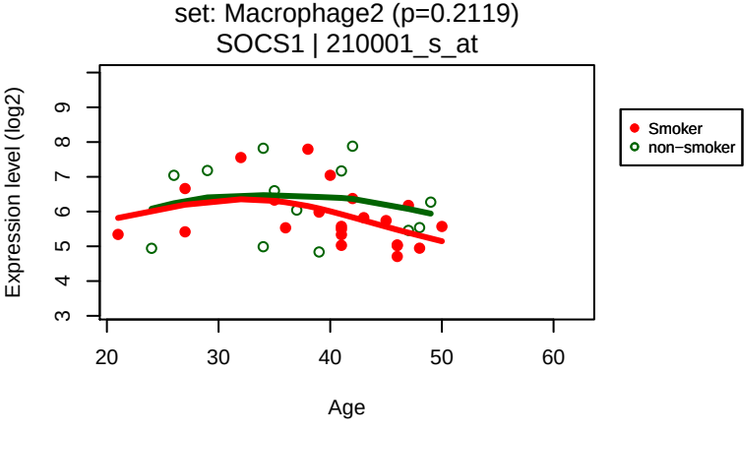

set: SmallAirway2 (p=0.2178)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker

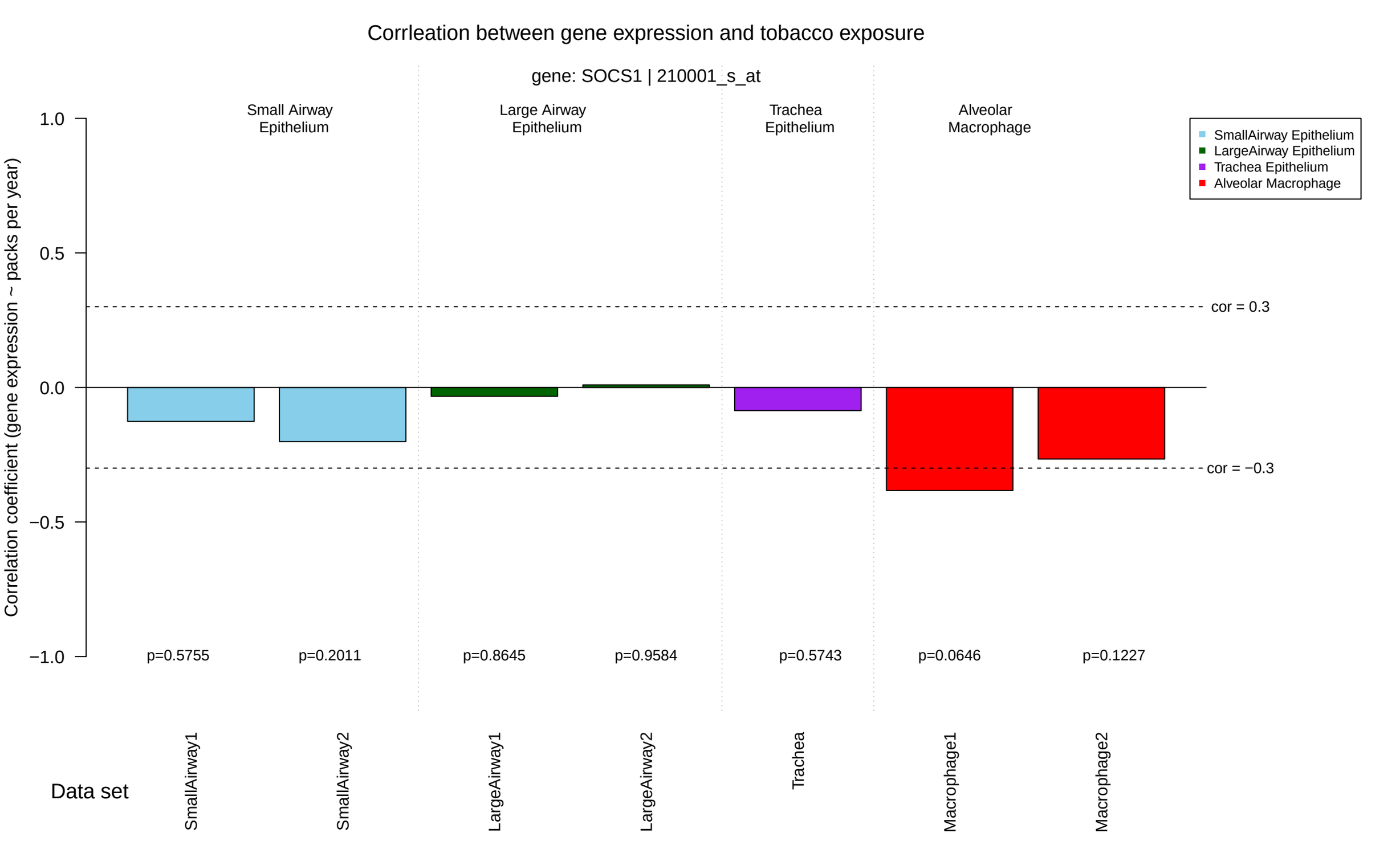

set: LargeAirway1 (p=0.9653)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.481)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4104)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0184)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2119)
SOCS1 | 210001_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS1 | 210001_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5755
p=0.2011
p=0.8645
p=0.9584
p=0.5743
p=0.0646
p=0.1227
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2